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Speech-based Parkinson's Disease Detection with Hybrid Machine Learning Approach

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***Abstract*—Parkinson's disease (PD) is a prominent neurodegenerative disorder that affects millions of people worldwide. It leads to irregularities in motor function, speech, and cognitive abilities. Early detection of PD is crucial, as addressing the disease in its initial stages can result in better management and slower progression. This study explores the application of ensemble machine learning (ML) methods to enhance the accuracy and reliability of early PD detection. The ensemble techniques—Bagging, Boosting, Voting, and Stacking—work by combining the strengths of multiple models to overcome the limitations of traditional approaches, thereby improving diagnostic performance. These methods were applied to a feature set of 11 features that includes vocal biomarkers (MDVP, jitter, shimmer, and HNR) etc, which reflect the effects of PD on voice. The combination of ensemble learning as a strong diagnostic tool provides an overall accuracy of 97%, which was achieved in our experiment using 195 vocal samples (147 PD and 48 healthy). The high accuracy levels attained here seem to point toward ensemble methods, potentially providing a clearer advance for PD detection over single deep learning models. In the future, these methods can be validated on larger and more heterogeneous datasets and methods from deep learning will be explored to refine the diagnostic model even further.**

***Index Terms*—Parkinson's disease, Speech Feature, Machine Learning, Ensemble Methods, Early detection, Vocal biomarkers**

I. Introduction

Parkinson's disease (PD) is a progressive brain disorder that affects motor and non-motor functions, including speech, cognition, and behavior. While PD primarily impacts older adults, millions worldwide suffer from this debilitating condition. The importance of early and accurate diagnosis cannot be overstated, as current clinical diagnostic methods rely heavily on observable motor symptoms, typically detected after significant neural damage has occurred [11]. This delay limits opportunities for early intervention and disease management, underscoring the need for more advanced diagnostic approaches.

In recent years, machine learning (ML) techniques have emerged as promising non-invasive methods for detecting PD by analyzing subtle early indicators such as handwriting patterns and speech features [3] [9]. Among these, vocal features—such as pitch, frequency, and voice modulation—are often among the earliest signs of PD. By leveraging these features, ML models have demonstrated the ability to predict PD with increased accuracy, enabling earlier and potentially more effective interventions.

However, traditional ML models, including decision trees and support vector machines (SVM), often face limitations in complex medical cases due to issues like overfitting and poor generalization [6], [12]. To address these challenges, ensemble learning techniques, such as Bagging, Boosting, Voting, and Stacking, combine multiple models to improve predictive performance, accuracy, and robustness. These methods are particularly valuable in scenarios involving diverse datasets and complex patterns, such as speech-based PD detection.

Several studies have explored the effectiveness of ensemble approaches for PD diagnosis. For instance, Fayyazifar et al. (2017) demonstrated that Adaboost and Bagging methods could significantly reduce overfitting and improve diagnostic accuracy [2]. Sheibani et al. achieved 90.1% accuracy using ensemble methods on voice data from 54 PD patients [10]. Similarly, Joshi et al. (2022) employed Stacking with KNN, SVM, and Random Forest (RF) models, achieving validation/test accuracies between 90% and 91% for vocal analysis [5].

Additional advancements have been reported in studies focusing on fine-tuned convolutional neural networks (CNNs). Hireš et al. (2022) ensembled multiple CNN models to analyze voice data from 50 PD patients and 50 healthy controls [4]. Ali et al. (2023) developed a novel EOFSC ensemble by integrating classifiers based on three vowel phonetics ('a,' 'u,' and 'o') [1]. In a separate study, Nilashi et al. (2022) combined clustering techniques with SOC and EM ensembles, denoising data using PCA, and predicting PD using SVM [8].

Given the sensitivity of speech features—particularly jitter, shimmer, and harmonics-to-noise ratio (HNR)—to early PD detection, this study builds on previous research by leveraging ensemble learning on vocal feature-based datasets [3]. The aim is to enhance diagnostic accuracy and contribute to the development of robust, generalizable models for early PD detection.

This research demonstrates the potential of ensemble methods in tackling the limitations of individual models, providing new insights for more effective machine-learning-based solutions for PD diagnosis.

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II. Methodology

A. System Design

The system design for the proposed PD detection framework consists of four interconnected stages. It begins with data collection, where voice samples are collected from patients, ensuring a diverse dataset that captures various demographics and severity of the disease while adhering to ethical guidelines. Following this, the feature extraction stage identifies relevant biomarkers, such as fundamental frequency, jitter, shimmer, amplitude variation, and noise-to-harmonics ratio, which are closely linked to speech impairments caused by PD. Advanced signal processing techniques, such as Mel-Frequency Cepstral Coefficients (MFCCs), are often employed to enhance the accuracy of feature representation. Once the features are extracted, ensemble classifiers are utilized in the classification stage to differentiate between PD-positive and PD-negative samples. Techniques like Bagging, Boosting, and Stacking are used to improve robustness, reduce bias, and enhance overall classification accuracy. Finally, the prediction output stage provides a probability score for each sample, reflecting the likelihood of PD. This output aids clinicians by supporting diagnostic decisions and enabling early and accurate identification of PD, ultimately contributing to better patient outcomes.

B. Dataset

This study uses the dataset with extracted PD and healthy control voice data features. There are 195 samples where 147 are with PD and 48 are healthy controls. These are collected from 31 peoples, 23 with PD. 10 Vocal biomarkers (MDVP (measures of fundamental frequency variation), jitter (frequency variation), shimmer (amplitude variation), Harmonics-to-Noise Ratio (HNR) etc., measuring various factors of vocal tremor and stability are among the key features in this dataset. These biomarkers were chosen based on their known significance as effects of PD on motor area control and motor output relating to voice. Dataset collection process is explained in details in the study conducted by Little et al. from where it was collected [7].

C. Data Preprocessing

These pre-processing steps were carried out precisely so that accurate model training is designed. Normalization was used to process the data collected in order to keep features on the same scale and not bias any of the models. Addressing missing values and performing feature selection, the data were prepared to be mapped by the models. Making these steps improved the performance of the model to find subtle vocal trend-lines associated with PD, which led to a more accurate prediction.

D. Model Architecture

We built several ensemble methods such as bagging, boosting, voting and stacking for the dataset. Then we also built deep learning models such as RNN and CNN model to compare the outcomes.

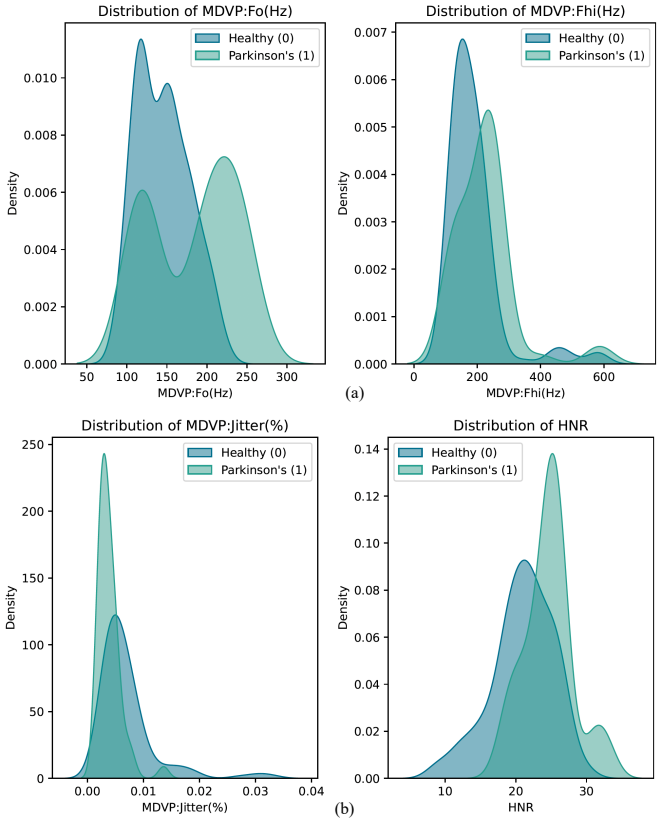

Fig. 1. Comparison of (a) MDVP variations (b) jitter and HNR variations in PD and Healthy Control

1) Bagging: A Random Forest(RF) method was deployed which used n_estimator of 100 i.e 100 decision trees (weak learners) of random subset of the dataset. The prediction outcome of the weak learners were summed to reduce the varience of the model.

2) Boosting: In Boosting ada boost was designed with max_depth set to 2. The n_estimator or the number of decision trees (Weak learners) was set to 150 and the learning rate is set to 0.5. Fig. 2 summerizes the model in a block diagram. The model trains the models sequentially, each trying to correct its predecessor. High correct rates have been established in medical data applications because this method focuses on error correction techniques.

3) Voting: For voting we used probability based soft voting technique with Logistic Regression, Support Vector Classifier(SVC), Decision tree, RF used in bagging and a KNN model was used. The KNN model was designed with k parameter 3. Decision Tree was tuned with min_sample_leaf 4 and mi_sample_split of 5. Th e SVC was designed with probabilty true, gamma parameter 10 and c parameter 1000. This technique trains all of the models independently and final prediction is made by majority of votes based on the probability of each models predicting Parkinson's and healthy controls. Since it combines the strengths of various types of model types with its Voting technique, it is robust too — since the errors of individual models are averaged out. Fig. 3 can

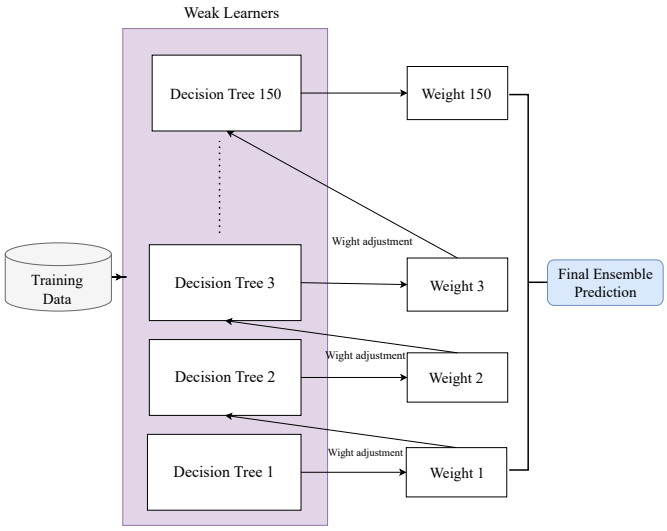

Fig. 2. Block Diagram of Adaboost Model used in this study

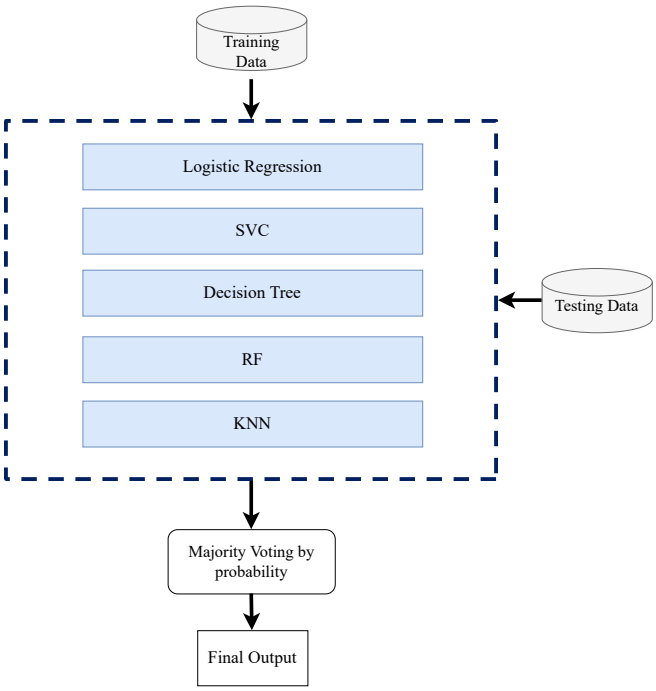

Fig. 3. Block Diagram of Voting method

be referred for voting method used in this study.

4) Stacking: Stacking uses a meta-model that is trained on the results of multiple base models. Through the combination of predictions through diverse base models, Stacking is likely to achieve a higher accuracy than any other single model can supply. In this work, Stacking is applied with a meta-learner which is logistic regression that integrates the strengths of the base models. The base models used are the Ada-boost used in boosting, RF model from bagging and previously designed SVC. All of these methods were implemented with Scikit-Learn libraries in Python. The model performance was evaluated using cross-validation with accuracy, precision, recall, and F1-score as the key metrics. These metrics were informative and can flow over the whole dimension of both models' performance and reliability for PD detection in potential clinical applications.

5) Deep Learning Models: An LSTM RNN model and a CNN model were used to compare the ensemble methods. The LSTM was designed with 64 hidden layer units and 1 fully connected output neuron. Activation function applied in the hidden layer is hyperbolic tangent and in the output layer it was sigmoid function. Binary cross entropy as loss function is used for the binary classification and Adam optimizer was used with learning rate of 0.001. To train the LSTM epoch number was taken as 50 and the batch size 16 was chosen to ensure stability with mini batch gradient update. The CNN architecture comprises a feed-forwarded NN comprising 2 convolutional layers, and two dense layers. The first convolutional layer has filter size of 32, kernel size of 3, ReLU as activation function. After that a maxpooling layer and a 25% dropout layer was used. Dropout is necessary to prevent over-fitting and ensure generalization. But 25% or lower was preferred as the dataset is small. Second convolutional layer is similar to the first one followed by similar pooling and dropout. only filter size was changed to 64. Then the first dense layer after flattening the multidimensional data comprises 128 neurons with activation function ReLU. After this another dropout of 20% was employed. The output layer has two fully connected neuron representing two classes: PD and Healthy controls. Softmax activation was applied in the output layer. Using tensorflow (Keras) in Python, both CNN and LSTM was established.

E. Dataflow Diagram

The dataflow diagram (Fig. 4) summarizes the steps for detecting PD using our ensemble ML model.

Voice samples are collected, preprocessed, and features are extracted. The ensemble classifiers and the deep learning models then process these input vectors to predict PD probability.

III. Results and Findings

Bagging, Boosting, Voting, and Stacking were employed, and each of these methods was evaluated based on metrics including accuracy, precision, recall, and F1-score. These metrics provide insights into the reliability of the model predictions and the feasibility of using them in clinical settings for detecting Parkinson's disease (PD). The predicted labels were compared with the actual diagnosis in the dataset to evaluate each method's performance, which reflects the model's effectiveness.

A. Performance Metrics

- **Accuracy**: The accuracy of a model reflects its performance as the ratio of true to false predictions for the

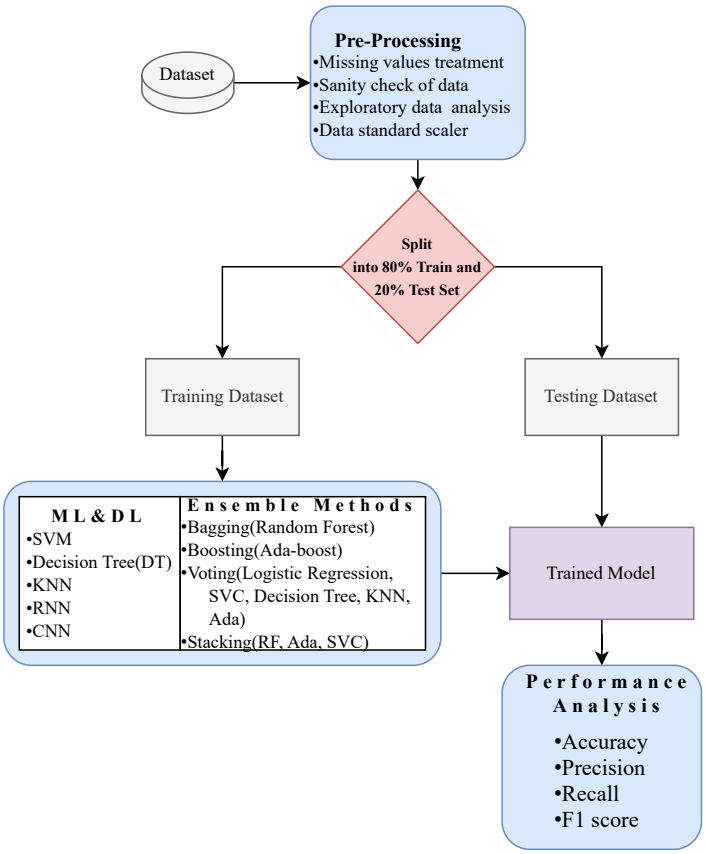

Fig. 4. Dataflow diagram for PD detection using ensemble ML model

classes relative to total predictions.. Accuracy was a key metric for the PD vs. non-PD classification task in this study, with the boosting methods achieving the highest accuracy at 97%. The voting and stacking yielded 95% accuracy. The RF model yielded 93% accuracy. The individual KNN model also obtained 92% accuracy all over the two classes. In comparison, the LSTM model obtained 92% and the CNN also yielded an accuracy of 92%

- **Precision**: Precision indicates the proportion of predicted positive cases that are actual positives. In PD detection, high precision reduces false positives, thus minimizing unnecessary testing and anxiety. High precision of all over .95 in ensemble methods signifies a low risk of false positives.
- **Recall**: Recall (sensitivity) is the proportion of actual positive cases (PD cases) identified by the model. This is crucial in clinical practice since missed PD cases have serious implications. The higher recall for Stacking, voting and Boosting models suggests they effectively detected true PD cases. All the ensembles yielded over .95 recall, but the CNN model resulted in .90 recall.
- **F1 Score**: The F1 score balances precision and recall, indicating the model's ability to detect PD accurately without favoring one metric over another. The Stacking model achieved the highest F1-score, highlighting its robustness for real-world applications.

B. Ensemble Model Comparison

Table I summarizes the classification reports for each ensemble methods and the DL models (RNN and CNN)

TABLE I
CLASSIFICATION REPORTS OF THE MODELS

Models	Precision	Recall	F1 Score	Accuracy(%)
Bagging	0.93	0.93	0.93	93
Boosting	0.97	0.97	0.97	97
Voting	0.95	0.95	0.95	95
Stacking	0.96	0.95	0.95	95
RNN	-	-	-	92
CNN	-	-	-	92

The receiver operating characteristics(ROC) curve curve of Voting and Adaboost models are depicted in Fig. 5 and Fig. 7. Roc represent these model's ability to distinguish the classes. For both the voting and boosting ROC curve area under the curve (AUC) is found to be 0.99.

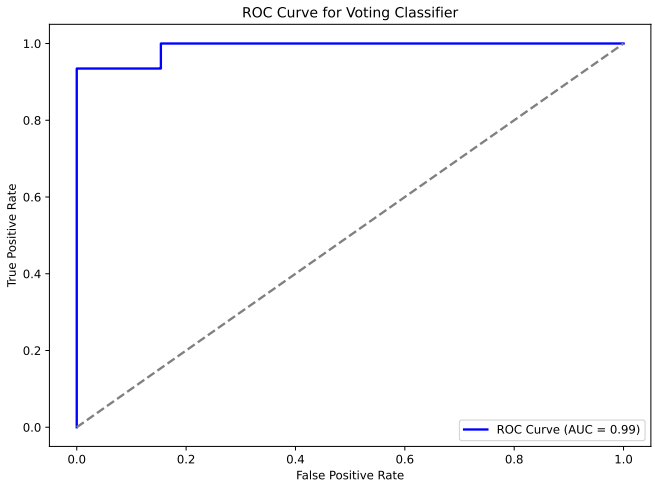

Fig. 5. ROC curve of Voting Method

The confusion matrix for the Stacking method, which is one of the ensemblers that achieved high accuracy, is shown in Fig. 6. The model can predict true PD data without any false predictions. But for true healthy control data, 3 false predictions were made. Other models also demonstrate high true prediction rate for both classes with a few false predictions for healthy control data

C. Interpretation of Results

The AdaBoost ensemble method achieved the best performance in terms of both accuracy and F1-score, offering the most balanced results for PD detection. Voting and Stacking

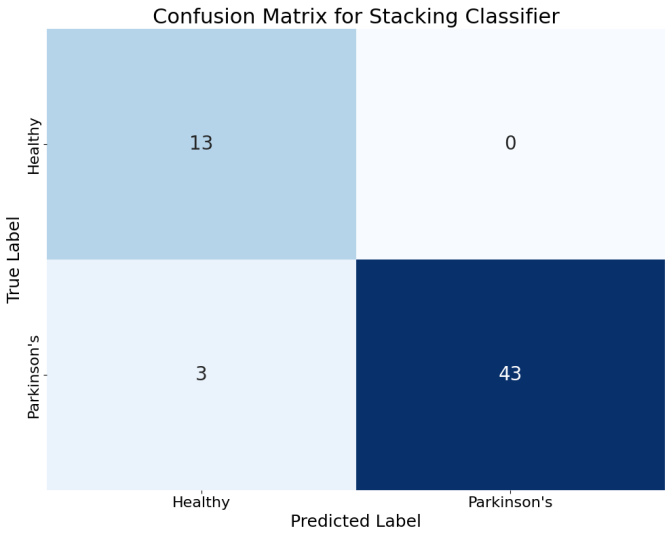

Fig. 6. Confusion Matrix for Stacking Method: for PD data, 43 true prediction, 0 false prediction; for healthy control data, 13 true and 3 false predictions

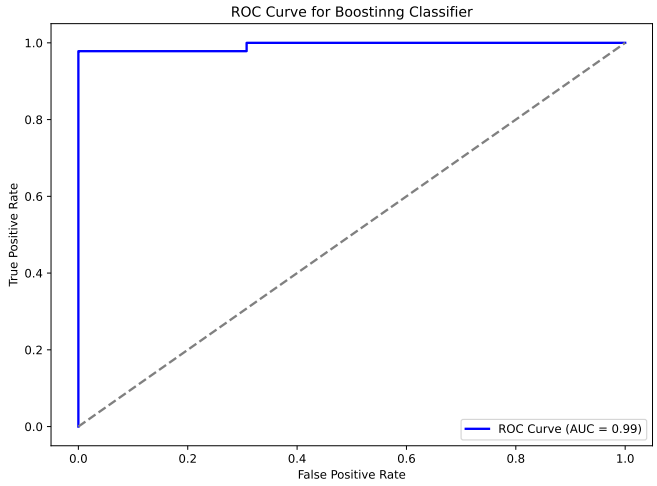

Fig. 7. ROC curve of Adaboost

methods also produced competitive outcomes, particularly excelling in precision and recall, highlighting their potential viability for PD diagnosis. Although Bagging showed lower accuracy compared to other ensemble methods, it demonstrated strong generalization by reducing variance, making it less prone to overfitting despite being less suited to the complexity of PD detection. The high AUC values of 0.99 for Boosting and Voting further indicate the reliability of these models in distinguishing true positives from false positives. However, the small, tabular, and non-sequential nature of the dataset may have affected model performance. Analysis of the confusion matrix revealed false predictions for the healthy control class, likely due to insufficient data, which may have led to overfitting during training.

While CNN and LSTM models also achieved high accuracy, the dataset size limited their generalization capacity. Additionally, tabular data derived from extracted audio features may not be suitable for these models. Future research should consider using time-series audio data with LSTM models and spectrogram representations with CNNs to capture sequential and spatial information for more robust classification.

The foundational study by Little et al. achieved a 92% classification accuracy using an SVM model with features such as HNR, RPDE, DFA, and PPE. While SVM requires careful feature scaling, tree-based ensemble methods have the advantage of automatically selecting the most important features for classification, enhancing their adaptability and performance in PD detection tasks.

D. Model Reliability

The results confirm that ensemble methods enhance the effectiveness of PD detection over conventional ML models while using feature extracted data. High values for accuracy, recall, and precision suggest that ensemble methods can serve as reliable clinical tools for early PD detection, aiding clinicians in managing PD progression effectively.Althogh a high acuracy result was obtained through this study, the result was overwhelmed with 75% PD data.But future research is needed with more diverse and larger data. Further research could explore the reliability of individual features in predicting Parkinson's disease.

IV. Conclusion

Parkinson's disease, the most common neurodegenerative disorder, often causes voice impairments in patients, making early and accurate diagnosis crucial for effective disease management. This study explored several ensemble machine learning (ML) methods, including Bagging, Boosting, Voting, and Stacking, for the early detection of PD using vocal biomarkers. Compared with traditional single-model approaches, ensemble methods demonstrated superior diagnostic accuracy and robustness, with the Boosting method achieving a notable 97% accuracy. These models effectively leveraged vocal features indicative of PD symptoms, combining the strengths of multiple algorithms to produce robust predictions and contribute to broader clinical classifications. However, to improve generalizability, future research should focus on validating these models using larger and more diverse datasets. Integrating deep learning techniques with ensemble methods may further enhance diagnostic accuracy, particularly by incorporating time-series audio data and audio spectrograms to capture sequential and spatial information. Such advancements hold the potential to develop precise diagnostic techniques, facilitating early detection, timely intervention, and better disease management for PD patients.

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